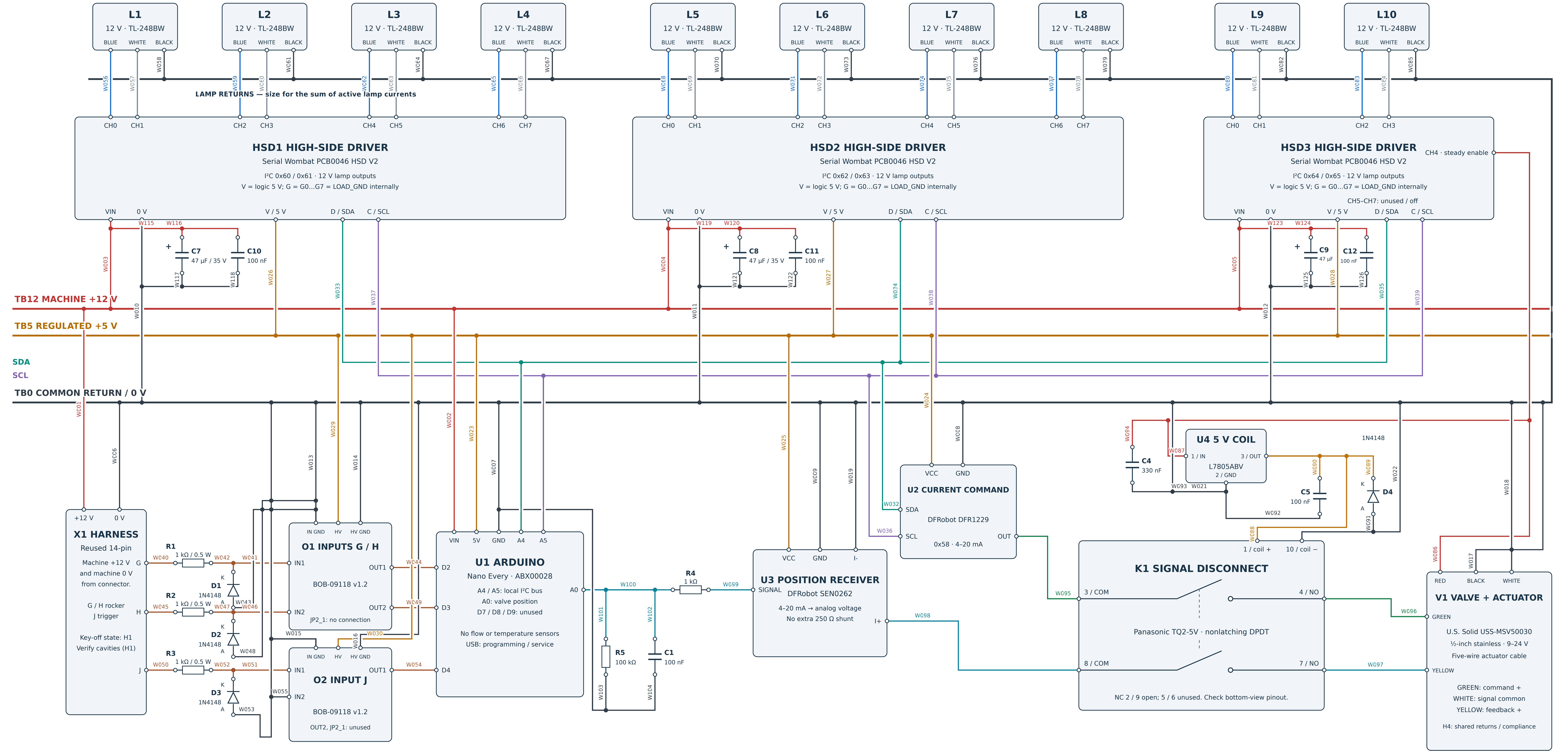


Read by terminal name, not by physical pin position.

Solid dot = joined wires. Gapped crossing = no connection. W-numbers match the connection register.

BENCH DESIGN — H1-H5 QUALIFICATION STILL REQUIRED

X1 labels are project signals, NOT the machine's numbered connector cavities.



CONTROL WITHOUT A FLOW METER

Valve command and position feedback remain connected.

Normal, Set Max, Flush and all ten lamps retain their controls.

An optional bucket / stopwatch calibration can map levels to flow.

A stored curve estimates flow only under similar water conditions.

No automatic flow sweep, live GPM or water-stop detection is provided.

Use the detailed booklet for valve-position qualification and recovery.

Live G/H/J with main power OFF: opto inputs draw about 9 mA each at 12 V.

Output-side HV = controller logic 5 V only. Bench-check the power-off case.

BUILD AND READING NOTES

- All 114 external wires are drawn. Shared trunks are joined terminals, not additional electronics.
- Component terminal positions are arranged for readability. Use the detailed sheets for physical pinouts.
- U1 GND1 = GND2; U2 GND = OUT_GND; U4 pin 2 = tab; HSD grounds are internally common.
- I²C stays inside the enclosure at 100 kHz. Use only the U2 SDA/SCL pull-ups; open all HSD bus pull-up jumpers.
- HSD1 address jumpers open; HSD2 A1 closed; HSD3 A2 closed. All MAX_AMP jumpers open.
- HSD3 CH4 is steady valve/relay enable, reserved for the proposed watchdog. Never include it in lamp effects.
- Resistors: 1%. Ceramic capacitors: X7R, 50 V. C7–C9: 47 μF, 35 V, 105 °C; observe polarity.
- Unused Nano pins: 3V3, A1/A2/A3/A6/A7, AREF, RESET1/2, RX/TX, D5/D6/D7/D8/D9/D10/D11/D12/D13.
- No flow meter or external temperature sensors. FM1, R6–R9 and C2 are removed; wire IDs are retained.
- Plumbing: hose → shutoff / strainer → valve → sprayer. Support the plumbing and keep electrical wiring dry.

Power loss does not spring-return the ball valve closed. Keep an accessible manual water shutoff.